Uncovering Historical Insights: A Framework for Explaining Historical Data through Graphs and LLM

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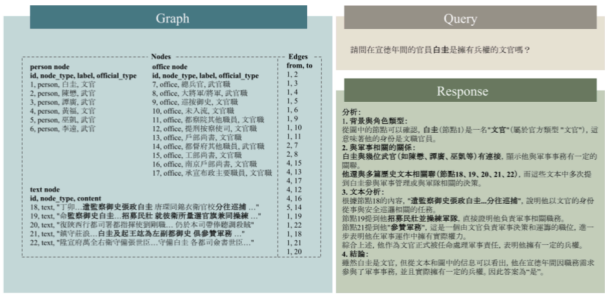
Introduction

Historical research highlights the critical role of social relationships and interactions in shaping historical contexts. By analyzing official roles and textual descriptions in historical records, historians can construct relationship network graphs that elucidate power dynamics, social connections, and interaction patterns.

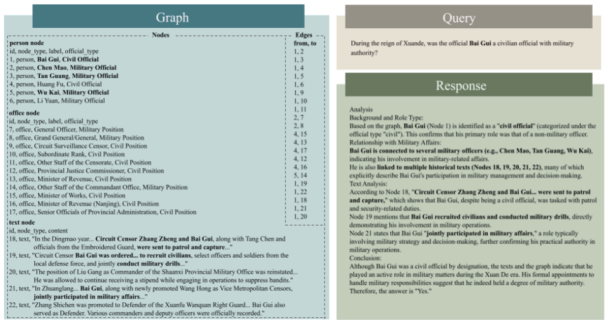
In this study, we propose a novel framework for explainable historical network analysis, leveraging relationship graphs to enhance knowledge retrieval and aggregation with the support of Large Language Models (LLMs). This approach provides transparent insights into historical power structures and relationships, offering researchers a robust tool to interpret complex historical narratives with clarity.

Focusing on the Ming Dynasty, a period characterized by the formal separation of civil and military domains, this study examines the nuanced interplay between these spheres. While the civil-military dichotomy often oversimplified the underlying power dynamics, our methodology uncovers hidden connections and interactions. By utilizing a constructed network graph of Ming officials and their roles (Chen, et al., 2023) in conjunction with textual records of military activities (Chen, et al., 2023) , we reveal the intricate relationships that defined this era.

To illustrate, we present an example from our graph-based Retrieval-Augmented Generation (RAG) system, as shown in Figure 1 and Figure 2. The system identifies Bai Gui as primarily associated with civil positions but highlights his strong connections to multiple military officials, suggesting involvement in military affairs. By integrating these connections with descriptions from military-related texts, the system demonstrates that Bai Gui possessed a degree of military authority. The LLM generates clear, comprehensible natural language responses that justify these conclusions, providing users with both interpretability and accessibility to complex historical data. Unlike traditional RAG systems that primarily rely on semantic similarity, our framework utilizes relationship graphs to explicitly model connections and derive nuanced insights embedded in historical narratives. This method enhances the transparency of historical analysis and mitigates the 'black box' limitations of conventional deep learning approaches. Researchers can trace and understand intricate historical relationships and dynamics with greater precision, offering a transformative tool for the field of digital humanities.



Bai Gui's graph-based analysis.



Bai Gui's translated graph-based analysis.

Dataset

Graph Data

We utilize the MingOfficial dataset (Chen, et al., 2023) , which details Ming Dynasty officials' biographical and professional records from the IHP database. Officials are represented as nodes, with edges denoting hierarchical ties, co-occurrences in roles, and shared service under specific reign titles. To analyze civil officials' military authority, military-related passages from the Ming Shi Lu (明實錄) provided by Academia Sinica (2013) were extracted using keyword matching and context analysis, then linked to officials via named entity recognition and temporal alignment.

Query Data

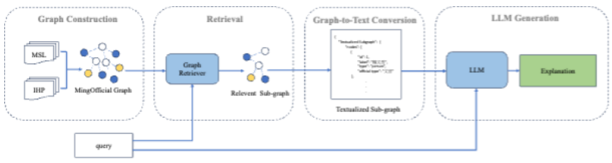
Based on the graph data, 438 queries were generated by extracting officials and their corresponding reign titles. These queries cover various historical periods and administrative levels, enabling targeted retrieval of subgraphs and textual data for detailed analysis of officials' roles and potential military involvement, as shown in Figure 3.

請問在 {執政皇帝年號} 年間的官員 {官員名稱} 是擁有兵權的文官嗎?
During the reign of {Emperor's Reign Title}, was the official {Official's Name} a civilian official with military authority?

Query template of our system.

Methodology

To address hallucinations and the context window limitations of LLMs, we adopt a RAG framework (He et al., 2024) . This framework bridges the gap between graph-based data and textual outputs by combining the retrieval of relevant graph segments with the generative reasoning of LLMs. The system consists of four components: graph construction, retrieval, graph-to-text conversion, and LLM-based generation, as depicted in Figure 4.



Overview of our system.

Graph Construction

We utilize a heterogeneous graph constructed from the MingOfficial dataset, designed to encode relationships between officials, positions, and military-related texts. This graph includes:

1. Nodes:
   - Person Nodes (P): Representing 8,721 officials, excluding royal family members and unrelated nobles, as they are outside the scope of military power analysis. Each node is enriched with embeddings generated using the metapath2vec model *(Dong et al., 2017).*
   - Office Nodes (O): Capturing all official titles, categorized as civil or military roles.
   - Text Nodes (T): Embedding military-related passages from the Ming Shi Lu using a BERT-based Chinese model (Wang et al., 2021).
2. Edges:
   - P-O Edges: Linking officials to their positions.
   - P-T Edges: Connecting officials to military-related textual passages.
   - P-P Edges: Dynamically updated during Graph Convolutional Network (Kipf et al., 2017) training. These edges are based on cosine similarity in embeddings, reflecting shared experiences inferred from textual data.
   - P-MT Edges: Connecting officials to military-related texts for targeted analysis.

This graph structure captures complex relationships between individuals, positions, and texts, forming a comprehensive database for downstream tasks.

Retrieval

Given the graph's size and complexity, a retrieval mechanism isolates query-relevant subgraphs to ensure computational efficiency and context preservation. The retrieval process consists of three steps:

1. Person Node Retrieval: Identifying the top-k officials most similar to the queried individual based on dynamically updated P-P edges.
2. Office Node Retrieval: Extracting positions linked to the retrieved officials via P-O edges.
3. Text Node Retrieval: Selecting military-related passages associated with the retrieved officials through P-MT edges.

By isolating relevant subgraphs, this step reduces noise while retaining the contextual data necessary for analysis.

Graph-to-Text Conversion

The retrieved subgraph is transformed into a textual representation for LLM reasoning. This textualization process encodes:

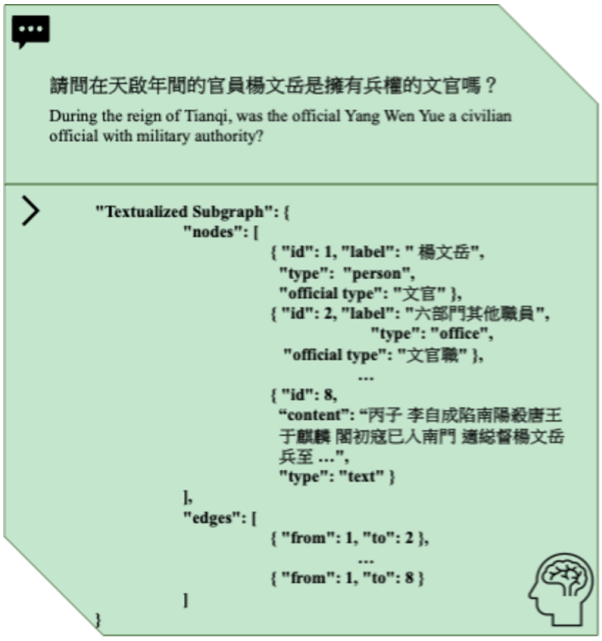
- Node Attributes: Including the official's name and role classification (civil or military).
- Relationships: Captured using "from" and "to" attributes that define edges between nodes.

The converted subgraph is stored in JSON format, ensuring compatibility with LLM inputs and preserving semantic relationships for generative reasoning.

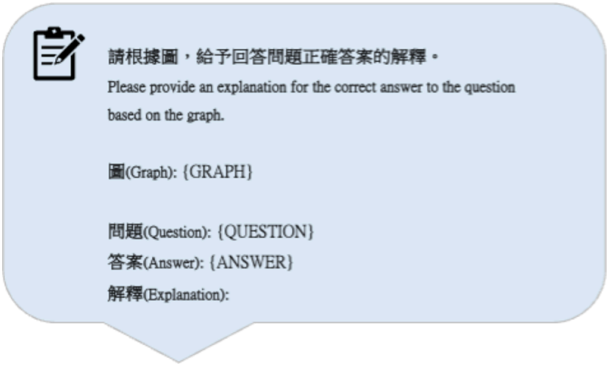
LLM-Based Generation

The system integrates user queries with the textualized subgraph into a structured prompt, as illustrated in Figure 5. This prompt enables the LLM to analyze relationships between nodes and edges, providing explanatory responses on whether the queried official held military power.

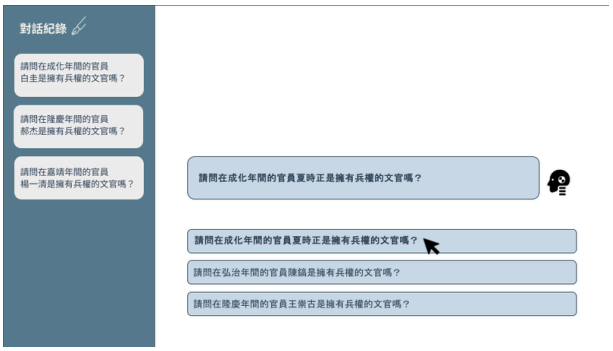
We use ChatGPT-4o-latest to enhance interpretability by generating explanations based on the textualized subgraph. The prompt structure, shown in Figure 6, guides the model to determine whether the official had military authority alongside civil duties or was exclusively a civil official. The system interface, as shown in Figure 7 and Figure 8, enables users to select predefined questions. Upon submission, the system retrieves the relevant subgraph, generates an explanation, and presents the output, as illustrated in Figure 9 and Figure 10, ensuring that the analysis is both interpretable and accessible to end-users.



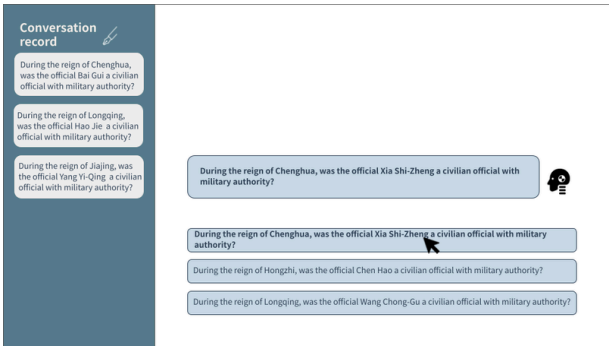
Textualized subgraph for Yang Wen Yue(楊文岳).



The prompt guiding the LLM's explanation.

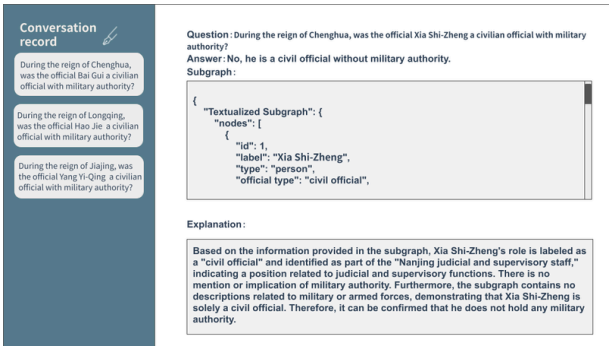


Users can choose questions provided by us.



Users can choose questions provided by us (translated version).

Our system will output the subgraph along with explanation.



Our system will output the subgraph along with explanation (translated version).

Experiments & Evaluation

To evaluate the effectiveness of our proposed explanation framework, we conducted experiments comparing three different input formats for generating explanations using LLMs. These formats were designed to assess the impact of providing structured reference information on the quality of generated explanations:

- Proposed Framework: Supplies a query-relevant subgraph, encoding structural relationships and textual data.
- Text Only: Provides only textual passages related to the query.
- No Reference: Offers no query-related reference information, relying solely on the LLM's pretrained knowledge.

Evaluation Criteria

The quality of generated explanations was assessed based on five key criteria:

- Relevance: The degree to which the explanation aligns with the query.
- Clarity: The ease with which the explanation can be understood.
- Consistency: Logical coherence and internal consistency of the explanation.
- Verifiability: The extent to which the explanation is supported by the provided reference data.
- Logical Structure: The organization and presentation of ideas within the explanation.

Explanations were scored on a scale of 1 to 10 for each criterion, with higher scores indicating better performance. The evaluation was conducted using GPT-based scoring, guided by the structured prompt shown in Figure 11 and Figure 12. Scores were averaged across multiple queries for each scenario.

Results

The evaluation results are summarized in Figure 13. Our proposed framework achieved the highest average scores across all five criteria, with particularly significant improvements in Verifiability and Logical Structure compared to the other scenarios. Key findings include:

- Proposed Framework: Consistently scored above 8.8 across all criteria, with a notable advantage in Verifiability, showing a 27.6% improvement over No Reference and an 8.15% improvement over Text Only. This indicates that the structured subgraph provided critical reference information, enabling the LLM to generate explanations grounded in specific relationships and textual evidence.
- Text Only: While producing moderately relevant and clear explanations, this scenario lacked consistency and verifiability due to the absence of relational context.
- No Reference: Explanations generated without reference information were often vague and less relevant, highlighting the limitations of relying solely on the LLM's pretrained knowledge.

根據問題、答案、解釋，給予解釋的評分：
問題：
"{graph}
{question}"

答案：{answer}
解釋：{explanation}

評分標準：
1. 相關性 (Relevance)
定義: 解釋是否直接回答問題，並圍繞主題展開。
衡量: 是否避免了與問題無關的資訊或冗餘內容？
衡量提問範例: 是否緊扣問題的核心？是否避免跑題？

2. 清晰性 (Clarity)
定義: 解釋是否語言簡潔、結構清晰，容易理解。
衡量: 是否避免了複雜術語或含糊其辭？是否依照邏輯順序表達？
衡量提問範例: 是否對目標受眾（如專家或非專業讀者）友善？是否有多餘的複雜句？

3. 一致性 (Consistency)
定義: 解釋是否在邏輯上前後一致，無自相矛盾之處。 衡量: 是否有邏輯跳躍或矛盾的結論？
衡量提問範例: 推理是否一致合理？是否能自洽？

4. 可驗證性 (Verifiability)
定義: 解釋是否可以透過參考文獻、數據或事實來驗證。
衡量: 是否提供了參考來源或明確提到依據？
衡量提問範例: 是否附有可靠的證據或來源？

5. 邏輯性 (Logical Structure)
定義: 解釋是否組織得當，有明確的開頭、主體和結論。
衡量: 是否條理分明、邏輯清晰，結論自然推導而出？ 衡量提問範例: 是否分步驟引導讀者理解複雜問題？

評分：以 1 到 10 的等級對以上10個維度進行評分，其中 1 表示遵守程度最低（差），10 表示遵守程度最高（優）。只要給每個維度評分的分數就好，不要多做解釋。

The prompt for explanation evaluation

Based on the question, answer, and explanation, provide a score for the explanation:
Question:
{graph}
{question}
Answer: {answer}
Explanation: {explanation}
Evaluation Criteria:
Relevance
Definition: Does the explanation directly answer the question and stay on topic?
Assessment: Does it avoid irrelevant information or redundant content?
Sample Assessment Questions: Does it focus on the core of the question? Does it avoid going off-topic?
Clarity
Definition: Is the explanation concise, clearly structured, and easy to understand?
Assessment: Does it avoid complex jargon or vague language? Is it presented in a logical order?
Sample Assessment Questions: Is it audience-friendly (e.g., for experts or non-experts)? Are there unnecessary complex sentences?
Consistency
Definition: Is the explanation logically consistent, without contradictions?
Assessment: Are there logical gaps or contradictory conclusions?
Sample Assessment Questions: Is the reasoning coherent and consistent? Does it hold up logically?
Verifiability
Definition: Can the explanation be verified through references, data, or facts?
Assessment: Does it provide references or clearly mention the basis for the explanation?
Sample Assessment Questions: Does it include reliable evidence or sources?
Logical Structure
Definition: Is the explanation well-organized with a clear introduction, body, and conclusion?
Assessment: Is it well-structured, logical, and does the conclusion follow naturally?
Sample Assessment Questions: Does it guide the reader step-by-step through a complex problem?
Scoring:
Rate each criterion on a scale of 1 to 10, where 1 indicates the lowest compliance (poor) and 10 indicates the highest compliance (excellent). Provide scores for each criterion only, without additional explanation.

The translated prompt for explanation evaluation

	Relevance	Clarity	Consistency	Verifiability	Logical Structure
Proposed Framework	**9.81**	**8.80**	**9.78**	**8.89**	**9.60**
Text Only	9.70	8.72	9.70	8.22	9.28
No Reference	9.39	8.41	9.53	6.97	8.82
Proposed Framework vs. Text Only (% Improvement)	1.13%	0.92%	0.82%	**8.15%**	3.45%
Proposed Framework vs. No Reference (% Improvement)	4.47%	4.63%	2.62%	**27.6%**	8.84%

The evaluation results

Conclusion

This study proposed a novel explainability framework integrating graph-structured data and LLM capabilities to enhance the analysis of historical datasets. By leveraging RAG techniques, the system effectively retrieves query-relevant sub-

graphs, and enable historians to analyze complex power structures and social interactions while providing clear, human-readable explanations.
Our experimental results validate the framework's effectiveness, with significant improvements in verifiability and logical structure compared to text-only and reference-free approaches. These findings demonstrate the value of integrating graph-based data with LLMs for producing accurate and substantiated explanations.
This framework offers a scalable solution for analyzing complex relationships and interpreting historical datasets, with potential applications in fields such as social history, economic networks, and cultural studies.

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